
Solution to Exercise Sheet 5:
Empirical distributions and learning

Foundations of Neuro-Symbolic AI
Summer term 2026, THWS, Alex Goessmann

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Task 1:

The empirical distribution $\mathbb{P}^D [X_0, X_1, X_2]$ is the average of the one-hot encodings of each observed state tuple.

- (i) We introduce a decomposition variable I taking states in $[10]$ enumerating the days. Based on this variable we find a CP decomposition

$$\mathbb{P}^D [X_0, X_1, X_2] = \langle \mathbb{P}[X_0|I], \mathbb{P}[X_1|I], \mathbb{P}[X_2|I], \mathbb{P}[I] \rangle_{[X_0, X_1, X_2]}$$

where $\mathbb{P}[I] = \frac{1}{10} \mathbb{I}[I]$ and the leg matrices are

$$\begin{aligned} \mathbb{P}[X_0|I] &= \begin{pmatrix} 1 & 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 & 1 & 1 \end{pmatrix} [X_0, I] \\ \mathbb{P}[X_1|I] &= \begin{pmatrix} 1 & 1 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 & 1 & 1 & 1 & 1 \end{pmatrix} [X_1, I] \\ \mathbb{P}[X_2|I] &= \begin{pmatrix} 1 & 0 & 1 & 0 & 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 1 & 0 & 0 & 1 & 1 \end{pmatrix} [X_2, I] . \end{aligned}$$

- (ii) We want to find the contraction of $\mathbb{P}^D [X_0, X_1, X_2]$ with the propositional formula

$$f [X_1, X_2] = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} [X_1, X_2] ,$$

representing the event that either both Alice and Bob bring an umbrella or both do not bring an umbrella. The rate of days on which this is true is the contraction

$$\langle \mathbb{P}^D [X_0, X_1, X_2], f [X_1, X_2] \rangle_{[\emptyset]}$$

We use the CP decomposition above and can ignore the core $\mathbb{P}[X_0|I]$ since X_0 (Rain) does not appear. Further, we notice, that $\mathbb{P}[I]$ only brings a factor of $\frac{1}{10}$. We are left with computing

$$\frac{1}{10} \cdot \langle \mathbb{P}[X_1|I], \mathbb{P}[X_2|I], f [X_1, X_2] \rangle_{[\emptyset]} = \frac{5}{10} = \frac{1}{2} .$$